पेटेंट कार्यालय
शासकीय जर्नल

OFFICIAL JOURNAL
OF
THE PATENT OFFICE

निर्गमन सं. 50/2025	शुक्रवार	दिनांक: 12/12/2025
ISSUE NO. 50/2025	FRIDAY	DATE: 12/12/2025

पेटेंट कार्यालय का एक प्रकाशन
PUBLICATION OF THE PATENT OFFICE

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2025
(21) Application No.202511104183 A
(43) Publication Date : 12/12/2025

(54) Title of the invention : VEHICLE TRAFFIC LIGHT DETECTION AND DRIVER ALERT SYSTEM

(51) International classification	:B60W0050140000, G06V0020580000, G08G0001095000, G08G0001096700, G06V0010820000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lovely Professional University
Address of Applicant :Lovely Professional University, Delhi-Jalandhar GT road Phagwara- 144411 Phagwara Punjab India
(72)**Name of Inventor :**
1)Akash Gupta
2)Alif Md Khairul Arshi
3)Sumit Kanchan

(57) Abstract :
VEHICLE TRAFFIC LIGHT DETECTION AND DRIVER ALERT SYSTEM The invention discloses a vehicle traffic light detection and driver alert system (100) comprising a rechargeable battery (102), a buck converter (104), an ESP32-CAM module (106), a relay module (108), a buzzer (110), and a traffic sign indicator (112). The ESP32-CAM (106) captures and processes images of the roadway to detect the color state of a traffic signal. Upon detecting a yellow signal, the ESP32-CAM (106) generates a control signal to activate the relay (108), which powers the buzzer (110) and traffic sign indicator (112), alerting the driver through audible and visual means. The system enhances safety by reducing signal violation due to driver distraction. FIG.1

No. of Pages : 14 No. of Claims : 10